# H ADITHYA KRISHNAN

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## Education

**Rajiv Gandhi Institute of Petroleum Technology** | **CGPA: 9.01** — *Jais, Uttar Pradesh*
*Bachelor of Technology in Computer Science and Design Engineering* — August 2024 - Present

**Indian Institute of Technology, Madras** | **CGPA: 8.63** — *Chennai, Tamil Nadu*
*Bachelor of Science in Data Science and Application* — September 2024 - Present

**St. Andrews Scots Senior Secondary School** | **85.6%** — *New Delhi, Delhi*
*Class XII (CBSE)* — 2024

**St. Andrews Scots Senior Secondary School** | **96.8%** — *New Delhi, Delhi*
*Class X (CBSE)* — 2022

## Experience

**Indian Institute of Technology (IIT) Delhi** — *New Delhi, Delhi*
*Research Intern, Department of Civil Engineering* — June 2026 - July 2026

- Developed the complete programmatic pipeline for the "Landslide Classification using Topological Data Analysis" project, replicating research that identifies landslide failure movements by analyzing their 3D shapes.
- Built data processing scripts to transform 2D landslide boundary polygons and Digital Elevation Models (DEM) into normalized 3D point clouds for spatial evaluation.
- Implemented Topological Data Analysis (TDA) algorithms to extract shape features—such as terrain compactness and slope variations—which act as proxies for physical landslide kinematics.
- Engineered and evaluated a Random Forest machine learning classifier to accurately categorize landslides into slides, flows, falls, and complex failures based entirely on their extracted topological properties.

## Projects

**EduAI: Multimodal RAG Pipeline** | Python, OpenAI Whisper, GPT-5/Llama 3.2, Ollama, Pandas

- Architected an end-to-end **Retrieval-Augmented Generation (RAG)** framework that automates video-to-audio conversion via **FFmpeg** and generates timestamped transcripts using **OpenAI Whisper (large-v2)**.
- Developed a custom **grouping algorithm** to merge short transcript segments into larger, logical blocks, providing the **LLM** with better context for more accurate answers.
- Implemented a **model-agnostic inference engine** supporting both high-performance **cloud APIs** (e.g., **GPT-5**) and private, local execution via **Ollama (Llama 3.2)**, ensuring **flexible deployment** across different hardware environments.
- Engineered a high-performance **vector search engine** using high-dimensional embeddings and Cosine Similarity to rank lecture segments, achieving **sub-second retrieval** from a structured Pandas DataFrame.

**Interactive Physics-Based Quadcopter Simulator** | Python, PyOpenGL, NumPy, Pygame

- Developed a real-time 3D flight simulator implementing **6-DOF** (Degrees of Freedom) rigid body physics and Newtonian mechanics.
- Implemented manual flight control systems (pitch, roll, yaw) using **NumPy** for vector arithmetic and **Euler angle**-to-matrix rotation transformations.
- Designed a textual Heads-Up Display (HUD) overlaying real-time telemetry data (velocity, g-force, gyro rates) onto the 3D viewport.

**California Housing Price Predictor** | Python, Scikit-Learn, XGBoost

- Orchestrated an end-to-end regression pipeline using **ColumnTransformer** for automated median imputation and one-hot encoding, utilizing **StratifiedShuffleSplit** to prevent sampling bias.
- Optimized model performance via **GridSearchCV**, comparing Linear Regression, Random Forests, and Decision Trees; diagnosed overfitting and selected **XGBoost** as the superior estimator.
- Achieved a **Test RMSE of 46,485** and an $R^2$ **Score of 0.84** with the final model, demonstrating significantly improved generalization over baseline predictors.

## Tech Stack

**Languages:** C, C++, Python, SQL.
**Libraries & Frameworks:** NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow, PyTorch, Streamlit, Giotto-TDA, GeoPandas, Rasterio.
**Databases:** MySQL, PostgreSQL.
**Tools & Visualization:** Power BI.
**Core Concepts:** Machine Learning & Deep Learning, Topological Data Analysis (TDA), Data Structures & Algorithms (DSA), OOPS, Database Management Systems (DBMS).

## Achievements & Activities

**Academic Excellence:** Ranked 2nd in the Computer Science and Design Engineering (CSDE) branch.
**Competitive Programming:** Solved over 595+ problems on Codeforces (Max rating: 1260).
**Positions of Responsibility:** Competitive Programming Co-Head, GeeksforGeeks Student Chapter.
**Extra Curriculars:** Winner, Intra-College Auction Football Tournament.